

Betas in Cost of Equity

Module 2 · CAPM, alternative risk measures & bottom-up beta

CAPM & MPT

Regression beta

Total beta

Other measures

Bottom-up beta

Which model?

1. CAPM & Modern Portfolio Theory

BETA

Beta (systematic risk) = sensitivity of the stock's return to the market return. It is the **non-diversifiable** risk.

$$K_e = R_f + \beta \times ERP$$

Modern Portfolio Theory (MPT)

- An investment should be evaluated by **how it affects the entire portfolio**.
- **Diversification can eliminate company-specific risk**.
- Investors require compensation **primarily for systematic risk**, so CAPM prices only systematic risk, through β (beta).

2. Historical (regression) beta

β is estimated by **regressing the company's historical returns against market returns**. The slope of the line is beta.

$$R_i = a + \beta R_m \text{ slope } (\beta) = \text{beta}$$

Why historical beta can be misused (pages 80-84)

	Problem	What goes wrong
A	High standard error	The estimated beta is statistically uncertain.
B	Low R^2	Shows how much of the stock's return variation is explained by market returns. Low means the market explains little.
C	Business mix changed	Beta may reflect the old business composition. The company may have moved into a different-risk business.
D	Financial leverage changed	Beta reflects the old capital structure. If D/E has changed materially, current equity risk can differ from the historical beta.
E	Exceptional market behaviour	Bubbles (dot-com), short squeezes, illiquidity or unusual trading give a mechanical beta number that is economically unhelpful. e.g. GameStop (pages 83-84).

3. Damodaran's alternative ways to measure risk

1 Relative volatility / Total beta

Total beta is a company-level scaling measure of **total volatility** relative to **market volatility**, so we use **standard deviation**.

$$\text{Relative volatility} = \text{SD}(\text{company}) / \text{SD}(\text{average stocks})$$

WHEN TO USE

It captures **all volatility**, **not just market-related**. Use it when you believe investors are **not perfectly diversified** and therefore want to consider company-specific risk.

	Normal CAPM beta	Total beta
Asks	Whether the stock moves relative to the market	How much the stock's return fluctuates w.r.t. market returns
Formula	$\beta_i = \text{Cov}(R_p, R_m) / \text{Var}(R_m)$	Total beta = σ_i / σ_m
Risk captured	Market-related risk only	Company + market risk (SD captures dispersion regardless of why)
How	Regression gives the value	Standard deviations of stock and market

EXAMPLE

Normal beta: market goes up 10% and the stock moves up 20% → $\beta = 2$.

Market returns: {+10%, -10%, +10%, -10%}

Stock A: {+15%, -15%, +15%, -15%}. It moves exactly with the market but **1.5x as much**.

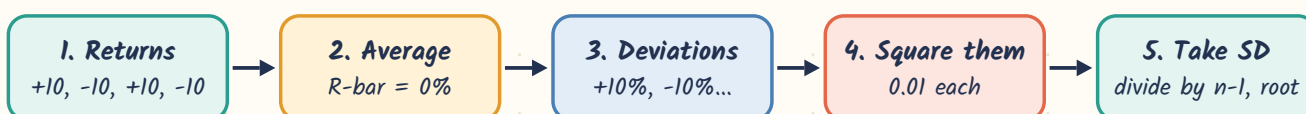
Stock B: {+15%, +5%, -20%, -5%}. Quite volatile, probably caused by **company-specific risk**, not market movements. Its σ may be high, but its covariance with the market may be low.

Volatility captures both company and market risk.

Volatility calculation

Suppose a stock's monthly prices are:

Month	Price	Return
Jan	100	-
Feb	110	$110/100 - 1 = +10\%$
Mar	99	$99/110 - 1 = -10\%$
Apr	108.9	$108.9/99 - 1 = +10\%$
May	98.01	$98.01/108.9 - 1 = -10\%$



$$\sigma = \sqrt{[\sum (R_t - R\text{-avg})^2 / (n - 1)]} \text{ (gives periodic volatility)}$$

EXAMPLE

$\sigma = \sqrt{[(0.01 + 0.01 + 0.01 + 0.01) / 3]} = 11.55\%$. This is **monthly** volatility because we used monthly returns.
 $\sigma(\text{annual}) = \sigma(\text{monthly}) \times \sqrt{12}$ $\sigma(\text{annual}) = \sigma(\text{daily}) \times \sqrt{252}$
 $11.55\% \times \sqrt{12} \approx 40\%$ annual volatility.

2 Proxy model

Instead of calculating beta directly, look at **characteristics of companies that seem to explain differences in risk / returns**. We notice that **small, highly leveraged and illiquid companies are riskier**. These become proxies for risk.

Company	Market cap	Debt / Equity	Liquidity	Risk score
A	1000 cr	0.2	High	Low
B	100 cr	1.5	Low	High

$$\text{Risk} = a + b_1(\text{Size}) + b_2(\text{Leverage}) + b_3(\text{Liquidity})$$

We are **not directly observing risk**. We use something related to risk as a substitute.

EXAMPLE

ABC Ltd was incorporated 6 months ago and listed as well, so a reliable historical beta is difficult. But we know it is a **small company with high debt, low liquidity and highly volatile earnings**. Compare it with thousands of comparable companies and **infer beta(ABC)** from those characteristics.

LIMITATION

Weakness: may not perfectly represent risk.

3 CAPM Plus

$$K_e = R_f + \beta \times \text{ERP} + \text{Additional premiums}$$

Keep CAPM, but **add premiums for risks that beta does not adequately capture**. e.g. a **country risk premium** if it is not captured in ERP.

4 Accounting-based risk measures

- A. Earnings volatility:** use variability in profit / earnings instead of stock-price volatility. e.g. 100, 105, 98, 103 is much more stable than 100, 200, 50, 250.
- B. Financial ratios:** assess risk using D/E, interest coverage, cash ratio, profitability, liquidity etc. e.g. D/E = 2.5x and interest coverage = 1.5x indicate **more financial risk** than D/E = 0.2x and interest coverage = 10x.
- C. Accounting beta:** a beta-like measure built from accounting performance instead of stock returns. Regress company earnings growth (%) vs market earnings growth (%) to get the accounting beta.
- D. Financial distress models:** composite scores such as the **Altman Z-score** combine accounting variables to indicate distress risk.

5 Qualitative risk models

Some risks cannot be captured by historical prices or ratios. Assess **management quality, competitive advantage, customer concentration, dependence on one product, regulatory exposure, industry conditions, governance** etc.

EXAMPLE

A company generates **80% of revenue from one customer** has major concentration risk, even if its current β and accounting ratios look normal.

This method usually gives **scores rather than a universal beta-equivalent formula**. It supports required-return **judgement** rather than mechanically replacing CAPM.

6 Debt-based risk measure

If the company's debt is traded or its borrowing cost is observable, the **lender's required return contains information about company risk**.

$$K_e = K_d \times \text{Scaling factor}$$

Equity is generally riskier than debt, so the cost is **scaled upwards**.

4. Bottom-up beta: the concepts

Traditional CAPM uses the **equity beta**, also called **levered beta**. It includes the effect of the company's financial leverage (debt).

UNLEVERED BETA

Unlevered beta: beta after removing the effect of financial leverage. It represents business risk but **still reflects operating leverage**.

Pure business risk



Plus operating leverage
= Unlevered beta



Plus financial leverage
= Levered beta

Term	Meaning
Pure business risk	Risk arising from the nature of the business: cyclical, discretionary vs essential products, competitive conditions, growth characteristics etc.
Operating leverage	Higher fixed costs generally increase unlevered beta, as risk increases.
Financial leverage	More debt generally means more risk borne by equity, which raises levered beta.

Business characteristic	Beta implication (pure business risk)	Reason
More cyclical	Higher	Depends on economic cycles / recession
More discretionary	Higher	Purchases postponed during recession
Luxury vs basic	Luxury higher	Demand falls more in recession
Higher growth expectations	Generally higher	Higher uncertainty

Operating leverage

A business with high fixed costs (F.C) can experience a much larger percentage change in operating profit for a given percentage change in revenue.

$$\beta_u = \beta_{\text{pure business}} \times [1 + F.C / V.C] \quad (\beta_u = \text{unlevered beta})$$

PRACTICAL LIMITATION

Companies generally **do not disclose a clean fixed / variable cost split**, so you cannot normally separate pure business beta in ordinary comparable-company valuation. We take unlevered beta **directly, by unlevering the levered beta** (see below).

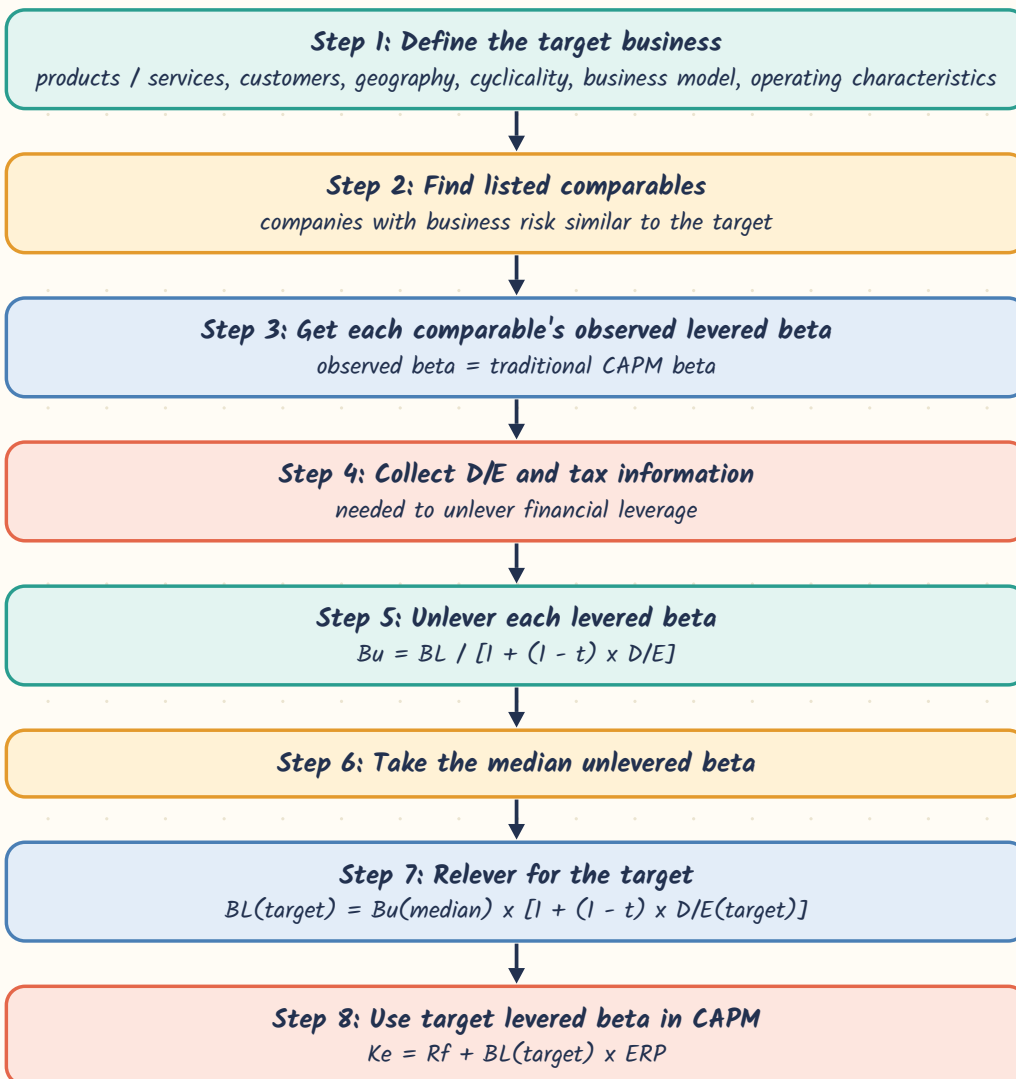
Financial leverage: levered beta

$$\beta_L = \beta_u \times [1 + (1 - t) \times D/E] \quad (\beta_L = \text{levered beta}, t = \text{tax rate})$$

IMP NOTE (V.V.IMP)

You will not find unlevered beta or pure business beta published anywhere, because it is **subjective** if estimated directly. That is why we take the **observed levered beta and unlever it**.

Bottom-up beta: the complete practical process



Step 2: what to compare (in priority order)

Priority	What to compare
1	Same core business
2	Same products / services
3	Same customer type
4	Similar geographic / economic exposure
5	Similar operating characteristics / F.C structure
6	Similar size / growth

NOTE

Do not exclude a comparable because of high / low D/E. It is irrelevant, since we unlever it anyway. D/E is not the primary exclusion criterion.

Roughly **5-10 strong comparables** are often preferable to 50 weak ones.

Find comparables in the target's **annual report / investor presentation** (competitors), **stock exchange industry classifications** and **financial databases**.

5. Historical beta vs bottom-up beta

	Historical beta	Bottom-up beta
Source	Target company's historical returns	Comparable companies
Requires public trading history	Yes	No, not for the target
Reflects old business mix	Potentially yes	Can reflect the current mix
Reflects target's current leverage	Only if history is representative	Yes, after relevering
Useful for private / new company	No, or poorly	Yes
Practical valuation usefulness	Easy / observable	Often stronger for valuation

Practical decision guide: which model to use?

Situation	Use
1. Public company with a long, stable, representative trading history	Historical beta , after checking standard error and whether business mix / leverage remain representative
2. Historical beta unreliable: business changed, leverage changed, or target is private / new	Bottom-up beta , if suitable comparables exist
3. Investor is explicitly undiversified and total risk is intended to be priced	Total beta / relative volatility
4. Beta is usable but a specific risk is not adequately captured	CAPM Plus
5. No useful market-price data	Accounting, debt-based or qualitative models. If comparables exist, bottom-up beta is directly useful for a DCF

How many companies do you need?

Model	Data needed
Historical beta	One target company + market / index data
Total beta	One target company + market volatility data
Accounting / qualitative / debt	Target company information, along with benchmarking
Proxy model	A broad sample is needed if you are estimating the proxy relationships from scratch. An analyst usually uses an established relationship / data.
Bottom-up beta	A set of comparables (discussed above)

6. Conventional vs debt-based approach

$$\text{Conventional: } \beta_L = \beta_u \times [1 + (1 - t) \times D/E]$$

This assumes $\beta(\text{debt}) = 0$: debt has no systematic risk, so all additional risk created by financial leverage falls on equity.

Debt-based approach: the conventional approach is not perfectly realistic, because **debt itself can have market (systematic) risk**. A company's bonds can become riskier when the economy deteriorates. So the formula becomes:

$$\beta_L = \beta_u \times [1 + (1 - t) \times D/E] - \beta_{\text{debt}} \times (1 - t) \times (D/E)$$

The last term is the **less additional risk borne by debt holders**.

V.IMP

We **do not usually use it**, because $\beta(\text{debt})$ is difficult to estimate. We take $\beta(\text{debt}) = 0$.

7. Bottom-up beta: actual process with an example

Step 1: identify the businesses

What businesses does my company operate in? e.g. **steel, logistics, fertilisers**.

Step 2: find publicly traded companies in each business

Find comparable companies and obtain their regression / equity betas. These are **levered betas**.

Company	Equity β (levered)
A	1.20
B	1.10
C	1.50
Average	$(1.20 + 1.10 + 1.50) / 3 = 1.27$

We **cannot stop here**. We can use the **median** approach, but for that we must first **unlever each company's beta** and then take the median. Simplified approach (with average beta):

$$\text{Unlevered beta} = \text{Average beta} / [1 + (1 - t) \times \text{Average D/E}]$$

This removes differences in financial leverage.

Step 3: adjust for operating leverage & product characteristics

WATCH OUT

Don't assume two comparable companies have exactly the same business risk just because they are in the same industry.

- **Operating leverage:** if the target has much lower fixed costs than the comparables, do not blindly use the unlevered beta from earlier steps.
- **Product characteristics:** if the target has a different product risk (cyclical, discretionary vs basic etc.), adjust for the differences.
- These adjustments are **numerically difficult**. The best solution is to **include comparables identical on these two factors** (fixed cost, cyclical etc.).
- e.g. Company A has higher F.C and a cyclical product vs Company B with lower F.C and a basic product. The unlevered β for A may be adjusted **upwards** for higher risk, but it can't be quantified.

Step 4: estimate how much value comes from each business

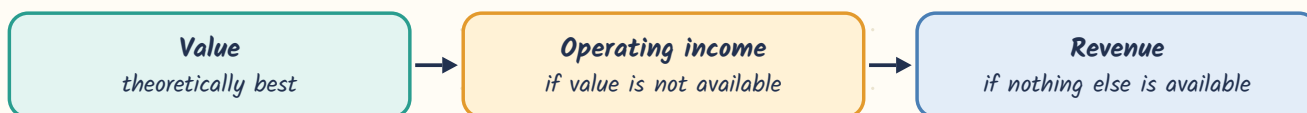
Business	Value	Weight
Steel	400 cr	$400/500 \times 100 = 80\%$
Logistics	100 cr	$100/500 \times 100 = 20\%$

NOTE

Why weight by value, not revenue / income? When calculating the beta of the entire company you are asking: how much of the company's economic value is exposed to each business risk?

Think of it as a portfolio: ₹20 lakh in Business A and ₹80 lakh in Business B means **80% of wealth is exposed to B**. It is illogical to weigh by "which one sells more". Weigh by **where my money / value is invested**.

Choose the weights in this order:



Operating income is generally better than revenue when businesses have **very different margins**.

Step 5: weighted average of unlevered betas

Steel $\beta_u = 0.80$ (weight 0.80). Logistics $\beta_u = 0.60$ (weight 0.20).

$$\beta_u = (0.80 \times 0.80) + (0.60 \times 0.20) = 0.76 \text{ (unlevered beta for the entire company)}$$

Step 6: relever for the target

Target D/E = 0.50, tax = 25%.

$$\beta_L = 0.76 \times [1 + (1 - 0.25) \times 0.50] = 0.76 \times 1.375 = 1.045$$

ANSWER

Put $\beta_L = 1.045$ into CAPM: $K_e = R_f + 1.045 \times ERP$

Can US and European aerospace companies estimate the beta of a Brazilian aerospace company?

*Potentially yes, because beta is fundamentally about **business risk**, and the underlying aerospace business can be similar across countries.*

BUT THERE ARE CONCERNS

- A. Different business characteristics
- B. Country risk
- C. Operating leverage
- D. Government exposure
- E. Capital structure